

Marco Giuliano D'Amico

Tech Lead · Full Stack Developer

Buenos Aires (CABA), Argentina · mgiulianodamico@gmail.com · github.com/GDeDev

ABOUT

Tech Lead and full stack TypeScript developer in fintech, working on Argentina's capital markets (broker-dealer / clearing agent).

I lead a development team at Vulcan Tech (Centaurus Group): market and back-office integrations, a P&L data lake on AWS, cloud security and IaC. I drive AI-assisted development across the team.

EXPERIENCE

Tech Lead · Full Stack Developer

2024 – Present

Vulcan Tech · Centaurus Group (broker-dealer)

Development of systems for a clearing and settlement agent, built from scratch on a microservices-oriented architecture. I lead the technical team (development, support and product), of up to 13 members, and set the platform's technical direction.

- Co-designed the platform from scratch with the infrastructure engineer: AWS accounts, Kubernetes clusters, replicas and environments (our own development setup and production inside the group's network). I define the systems architecture and validate every component before it reaches production.
- Integrations with markets, order management systems, financial data providers and external back-offices: paginated ingestion with retries, async jobs (SQS, Kubernetes workers, Redis pub/sub) for queries that exceeded the HTTP timeout, and migration of legacy consumers from .NET to Node.
- P&L data lake on AWS (S3, Parquet, Glue, Athena, Step Functions; RAW → CURATED layers) computing average cost, mark-to-market and realized / unrealized P&L.
- Cloud security and secure delivery: access through IAM Identity Center (scoped permission sets and MFA) and M2M authorization between APIs using scopes. Infrastructure as code with AWS CDK and GitHub Actions pipelines using OIDC, with vulnerability scanning (Trivy), pre-deploy secret validation and manual approval for production. PII anonymization in test environments.
- Internal developer platform (IDP) and in-house CLIs to manage Kubernetes, secrets, access and per-pod permissions. With templates and guides, a new API is deployed to 3 environments in ~10 minutes.
- Observability with Grafana, Prometheus and Loki: logs, API and database status, health checks and WAF, with email alerts so support can get ahead of incidents.
- Intelligent document extraction with generative AI (AWS Bedrock): an OCR-style service that reads documents in multiple formats and returns structured data, reusable for any document type.
- Central service documentation and AI adoption across the team: internal guides, spec-driven development and a Claude Code toolkit.

DevSecOps Consultant · Freelance

Jan – Jun 2026

Firmato

6-month contract alongside Vulcan Tech: technical audits of client systems.

- Code and architecture reviews: structure, maintainability and best practices.
- Assessed compliance with security standards, performance and CI/CD pipelines, with improvement recommendations.

Full Stack Developer · Frontend Instructor 2020 – 2024

NUCBA

"Galileo" submissions platform (2,500+ students) and institutional HUB, leading a team of 2 developers. Taught the Frontend program and ran JavaScript and React meetups.

- Designed and built the platform enrollment flow.
- Stack: React, Redux Toolkit, TypeScript, NestJS, Prisma, PostgreSQL.
- Migrated infrastructure from Digital Ocean to AWS: EC2/ECS, RDS PostgreSQL, ALB, ECR, Hasura and Strapi, with CI/CD on GitHub Actions.

Project Manager · Full Stack Developer 2020 – 2022

CJ Springs

End-to-end internal ERP for a spring manufacturer: stock, sales, cash flow and financial activity.

- Stack: TypeScript, Remix, Tailwind CSS, NestJS, Prisma, PostgreSQL.

Founder · Web development and content 2016 – 2021

QeoFamily

Automotive digital media brand with an Instagram community of 160K+ followers, built and operated independently.

- Brand website, built with HTML, CSS and JavaScript.
- Internal content management platform to publish and administer the site.
- Content, organic growth and campaigns with automotive brands.

PROJECTS

Socky

Cross-platform app (Android and iOS) to share the clipboard across devices, with its own backend.

React Native · Expo · Node · Express · Prisma · Docker

Bugsmith

Agent that reads a production error, finds the responsible code and proposes the fix in a PR for a person to review.

- How it works: retrieves repo context, diagnoses with the model, validates the fix deterministically (single search, syntax) and opens the PR. If there is no safe fix, it escalates to a person.
- Retrieval: lexical (BM25) or hybrid search with local embeddings, RRF fusion and reranking, compared with in-house evals.
- Two modes: fixed pipeline, or an agent that investigates with read-only tools (tool calling).
- Similar-incident memory and its own MCP server to query errors from Claude Code.

Python · FastAPI · SQLite FTS5 · fastembed · MCP · LLM evals · Next.js

Car photo organizer

Photo organizer for the automotive world: sorts large photo batches and drafts the text for posts and technical reports.

- Vision: a model tags each photo (make, model, angle, quality) with validated structured output. Deduplication and the quality filter use no AI.
- RAG over technical fact sheets (Wikipedia, Wikidata, NHTSA): chunking, hybrid BM25 + embeddings search with RRF and cited answers.
- Agents: a router hands off to the researcher (tool calling), the curator or the writer, and a verifier (LLM-as-judge) checks the facts against the sources.
- Several models with automatic fallback when a quota is hit, and a read-only MCP server.

Python · FastAPI · SQLite · Gemini · RAG · Agents · MCP · Next.js

STACK

languages	TypeScript · JavaScript · Python · SQL (T-SQL, Athena/Presto)
backend	NestJS · Node.js · FastAPI · Prisma · Jest · Hexagonal architecture · CQRS
frontend	Next.js · React · React Native / Expo · Tailwind CSS · shadcn/ui · TanStack Query · Zod
aws	Lambda · SNS/SQS · S3 · DynamoDB · ElastiCache · Bedrock · Athena · Glue · Step Functions · ECS/EC2 · RDS · IAM Identity Center · CDK
data	SQL Server · PostgreSQL · Redis · Parquet / data lake
devops	Docker · Kubernetes · GitHub Actions (OIDC) · Trivy · Turborepo · Doppler
observability	Grafana · Prometheus · Loki
ai	Claude Code (skills, subagents, plugins) · RAG (BM25 + embeddings, RRF, reranking) · Agents and tool calling · Vision models with structured output · LLM-as-judge and evals · MCP (client and server) · Guardrails and prompt-injection defense · Spec-driven development
domain	Capital markets (clearing, repos, P&L, reconciliations)

EDUCATION

nucba	Full Stack Program (2017 – 2019) · Backend Program · UX/UI Bootcamp
courses	Platzi · Crehana · Domestika: programming, UX/UI and data
languages	Spanish (native) · English (intermediate)